



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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CO 3
ASSESSMENT TOOL 2
Data Interpretation

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Course Name: Fundamentals of Data Science

1. Standardization/Normalization

A feature "Age" ranges from 18–75 and another feature "Income" ranges from ₹20,000–₹20,00,000 in a dataset meant for a distance-based algorithm. Explain, with reasoning (not just definitions), why failing to standardize these variables would distort analysis, and calculate the z-score for Age = 45 if mean = 40 and SD = 12.

Given:

- Age range = 18–75
- Income range = ₹20,000–₹20,00,000
- Age = 45
- Mean of Age = 40
- Standard Deviation = 12

Explanation:

In a distance-based algorithm such as KNN or K-Means, distances between data points are calculated using the feature values.

Age has a much smaller numerical scale than Income. For example, an Age difference of 10 has a numerical value of only 10, while an Income difference of ₹1,00,000 has a numerical value of 100,000. Therefore, without standardization, Income can dominate the distance calculation, even if Age is equally important.

Standardization converts the features to a comparable scale so that one feature does not dominate the analysis simply because it has larger numerical values.

Z-score Calculation:

Formula:

$$Z = (X - \text{Mean}) / \text{Standard Deviation}$$

Substitute the values:

$$Z = (45 - 40) / 12$$

$$Z = 5 / 12$$

$$Z = 0.4167$$

Answer:

$$Z\text{-score} = 0.417 \text{ approximately}$$

Therefore, Age = 45 is approximately 0.42 standard deviations above the mean.

2.Distribution Shape from Summary Stats Alone

A dataset has Mean = 45, Median = 45, Mode = 45, and skewness ≈ 0 , but kurtosis = 5.8 (leptokurtic). Explain what this combination tells you about the distribution's shape that skewness and central tendency alone would miss.

Given:

- Mean = 45
- Median = 45
- Mode = 45
- Skewness ≈ 0
- Kurtosis = 5.8

Solution:

Since:

Mean = Median = Mode = 45

the data is centered around 45.

The skewness is approximately 0, which indicates that the distribution is approximately symmetric. There is no strong tendency for the distribution to have a longer tail on either side. However, skewness alone cannot tell us about the peakedness and tail heaviness of the distribution.

The kurtosis is 5.8. Since this is greater than the normal-distribution kurtosis value of approximately 3, the distribution is leptokurtic.

A leptokurtic distribution generally has:

- A more concentrated/sharper center
- Heavier tails
- Greater possibility of extreme observations

Answer:

The distribution is approximately symmetric and centered at 45, but it is leptokurtic, meaning it has heavier tails and a greater tendency for extreme values compared with a normal distribution.

Thus, mean, median and skewness describe the center and symmetry, while kurtosis reveals information about the tails and peakedness that they cannot show.

3. Grouped Data — Mean and Variance

Class Interval: 0–10, 10–20, 20–30, 30–40, 40–50

Frequency: 5, 8, 15, 12, 10

Calculate the mean and variance using the step-deviation method. Show the assumed midpoints and working.

Given:

Class Interval	Frequency
0–10	5
10–20	8
20–30	15
30–40	12
40–50	10

Step 1: Calculate Class Midpoints

Formula:

Midpoint = (Lower Limit + Upper Limit) / 2

Class Interval	f	Midpoint (x)
0–10	5	5
10–20	8	15
20–30	15	25
30–40	12	35
40–50	10	45

Total frequency:

$$N = 5 + 8 + 15 + 12 + 10 = 50$$

Step 2: Use Step-Deviation Method

Take assumed mean:

$$A = 25$$

Class width:

$$h = 10$$

Formula:

$$d = (x - A) / h$$

Class	f	x	d	fd	fd ²
0–10	5	5	-2	-10	20
10–20	8	15	-1	-8	8
20–30	15	25	0	0	0
30–40	12	35	1	12	12
40–50	10	45	2	20	40
Total	50			14	80

Step 3: Calculate Mean

Formula:

$$\text{Mean} = A + h(\Sigma fd / N)$$

Substitute:

$$\text{Mean} = 25 + 10(14/50)$$

$$\text{Mean} = 25 + 2.8$$

$$\text{Mean} = 27.8$$

Therefore:

$$\text{Mean} = 27.8$$

Step 4: Calculate Variance

Formula:

$$\text{Variance} = h^2 [\Sigma fd^2/N - (\Sigma fd/N)^2]$$

Substitute:

$$\text{Variance} = 10^2 [(80/50) - (14/50)^2]$$

$$= 100 [1.6 - 0.0784]$$

$$= 100 \times 1.5216$$

$$= 152.16$$

Answer:

$$\text{Mean} = 27.8$$

$$\text{Variance} = 152.16$$

4. Comparing Two Distributions via CV

Portfolio A: Mean return = 12%, SD = 3%

Portfolio B: Mean return = 18%, SD = 5%

An investor claims Portfolio B is "riskier in absolute terms but not necessarily worse per unit of return." Evaluate this claim quantitatively using Coefficient of Variation and explain what CV captures that SD alone doesn't.

Given:

Portfolio A:

- Mean return = 12%
- Standard Deviation = 3%

Portfolio B:

- Mean return = 18%
- Standard Deviation = 5%

Formula:

$$CV = (\text{Standard Deviation} / \text{Mean}) \times 100$$

Portfolio A

$$CV = (3 / 12) \times 100$$

$$CV = 0.25 \times 100$$

$$CV = 25\%$$

Portfolio B

$$CV = (5 / 18) \times 100$$

$$CV = 0.2778 \times 100$$

$$CV = 27.78\%$$

Comparison

Portfolio	Mean	SD	CV
A	12%	3%	25%
B	18%	5%	27.78%

Interpretation:

Portfolio B has a higher standard deviation:

$$5\% > 3\%$$

Therefore, Portfolio B is **riskier in absolute terms**.

However, its CV is also higher:

27.78% > 25%

This means Portfolio B has slightly greater risk relative to its return.

Answer:

The investor's statement is partially correct.

Portfolio B is riskier in absolute terms because its SD is higher. However, according to CV, Portfolio B also has slightly higher variability per unit of return.

Therefore:

- **Portfolio A:** Lower relative risk
- **Portfolio B:** Higher return but higher relative risk

What CV captures:

Standard deviation measures absolute variability, whereas CV measures variability relative to the mean. Therefore, CV is useful when comparing datasets or investments with different average values.

15. Outlier's Effect on Variance

A dataset of 10 values has variance = 25. One value, originally 40 (close to the mean of 42), is mistakenly recorded as 400. Without recalculating exactly, explain directionally and reason about the magnitude of distortion to (a) the mean, (b) the variance — and why variance is affected disproportionately more than the mean.

Given:

- Number of observations = 10
- Original variance = 25
- Original value = 40
- Mean ≈ 42
- Incorrect value = 400

The value **40** is close to the mean of 42, but it is mistakenly changed to 400.

(a) Effect on Mean

The change in the value is:

$$400 - 40 = 360$$

Since there are 10 observations, the mean increases by:

$$360 / 10 = 36$$

If the original mean is 42:

$$\text{New mean} \approx 42 + 36$$

New mean ≈ 78

Therefore, the mean increases substantially.

(b) Effect on Variance

Variance is calculated using squared deviations:

$$\text{Variance} = \Sigma(X - \text{Mean})^2 / N$$

The original value 40 is close to the mean:

$$40 - 42 = -2$$

Squared deviation:

$$(-2)^2 = 4$$

But the incorrect value 400 is extremely far from the mean:

$$400 - 42 = 358$$

Squared deviation:

$$358^2 = 128,164$$

Therefore, the incorrect value contributes an extremely large squared deviation.

Why is variance affected more than the mean?

The mean uses the values directly. Changing one value from 40 to 400 increases the total by 360, so with 10 observations the mean increases by 36.

Variance, however, squares the deviation from the mean. Therefore, an extremely large value produces an extremely large squared deviation and can dramatically increase the variance.

Answer:

(a) Mean: The mean increases substantially, approximately from 42 to 78.

(b) Variance: The variance increases dramatically because the erroneous value 400 is extremely far from the other observations and variance uses squared deviations.

Conclusion:

A single extreme value can have a much stronger effect on variance than on the mean. This demonstrates why detecting and correcting outliers and data-entry errors is important before performing statistical analysis.